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Факультет программной инженерии и компьютерной техники

Итерация №3 по рефакторингу курсовой работы за 3 курс по предмету “Информационные системы”

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Этап 3. Оптимизация использования индексов

Цель: Проанализировать использование индексов на большом объеме данных (минимум 1 Гб), проверить действительно ли есть прирост в производительности от их использования и на основе проведенного анализа удалить ненужные индексы и оставить/добавить нужные.

Задачи:

1. Написание скрипта SQL, который генерирует большой объем тестовых данных для всех таблиц в базе данных.
2. Написать скрипты, которые позволяют оценить использование индексов, после того как база данных заполнена сгенерированными данными.
3. Последовательно проанализировать каждый используемый индекс, изменить его тип при необходимости, удалить в случае если он не дает значимого прироста в производительности.
4. Проверить корректность внесенных изменений, составив результирующую таблицу для всех используемых индексов.

Демо проекта

Ссылка на демо итерации рефакторинга — https://rutube.ru/video/private/83cc8e8f7c96084377bd33a66b36e7bb/?p=tINTrptRYM-eIUhWCwRS9g

Репозиторий проекта до начала рефакторинга — https://github.com/stoneshik/dar
Репозиторий проекта после исправлений данной итерации — https://github.com/stoneshik/dar/tree/third
Diff проекта изменений за итерацию — https://github.com/stoneshik/dar/compare/second...third

Описание изменений

Список индексов до оптимизации:

```
CREATE INDEX user_email_index_hash ON users USING hash(user_email);
CREATE INDEX user_login_index_hash ON users USING hash(user_login);

CREATE INDEX order_type_index_hash ON orders USING hash(order_type);
CREATE INDEX order_status_index_hash ON orders USING hash(order_status);

CREATE INDEX vending_point_schedule_day_week_index_hash ON vending_point_schedules
  USING hash(vending_point_schedule_day_week);
CREATE INDEX vending_point_unusual_schedule_date_index_hash ON vending_point_unusual_schedules
  USING hash(vending_point_unusual_schedule_date);

CREATE INDEX function_variant_index_hash ON function_variants USING hash(function_variant);

CREATE INDEX machine_supplies_datetime_index_btree ON machine_supplies USING
btree(machine_supplies_datetime);
CREATE INDEX machine_condition_datetime_index_btree ON machine_conditions USING
btree(machine_condition_datetime);
```

Для анализа индексов, был написан скрипт который генерирует большой объем данных (более 1 Гб).

Проверка объема данных в заполненной БД:
-- Размер базы данных
SELECT pg_size_pretty(pg_database_size(current_database())) as database_size_bytes;

is: -- Размер базы д...

database_size_...
1569 MB

Также был реализован скрипт проверки индексов, который сначала выполняет запросы с отключенными индексами, и затем запросы с включенными индексами. Для того показывалась нужная для анализа информация ставится EXPLAIN (ANALYZE, BUFFERS) перед телом SELECT запроса — таким, образом виден план выполнения запроса, вместе со временем и затраченными ресурсами.

1) Проверка user_email_index_hash:
Запуск запроса без индекса:

is: multiple query results

SET enable_bitmapscan = off; | -- 1. not_index user_email_index_hash E... | -- 2. not_index user_login_index_hash EX... | -- 3. not_index ord

QUERY PLAN
Aggregate (cost=11592.75..11592.76 rows=1 width=8) (actual time=82.863..82.865 rows=1 loops=1)
Buffers: shared read=10660
-> Seq Scan on users (cost=0.00..11592.75 rows=1 width=0) (actual time=82.857..82.858 rows=0 loops=1)
Filter: ((user_email)::text = 'user_50000@example.com'::text)
Buffers: shared read=10660
Planning:
Buffers: shared hit=109 read=29
Planning Time: 0.500 ms
Execution Time: 82.915 ms

Запуск запроса с индексом:

is: multiple query results

SET enable_bitmapscan = on; | -- 1. index user_email_index_hash EXPLA... | -- 2. index user_login_index_hash EXPLAI... | -- 3. index order_type_index_ha

QUERY PLAN
Aggregate (cost=8.02..8.03 rows=1 width=8) (actual time=0.037..0.037 rows=1 loops=1)
Buffers: shared read=3
-> Index Scan using user_email_index_hash on users (cost=0.00..8.02 rows=1 width=0) (actual time=0.034..0.034 rows=0 loops=1)
Index Cond: ((user_email)::text = 'user_50000@example.com'::text)
Buffers: shared read=3
Planning Time: 0.060 ms
Execution Time: 0.054 ms

2) Проверка user_login_index_hash:

Запуск запроса без индекса:

is: multiple query results

ексов - ... | SET enable_bitmapscan = off; | -- 1. not_index user_email_index_hash E... | -- 2. not_index user_login_index_hash EX... | -- 3. not_index ord

QUERY PLAN
Filter...
Aggregate (cost=11592.75..11592.76 rows=1 width=8) (actual time=82.831..82.833 rows=1 loops=1)
Buffers: shared read=10660
-> Seq Scan on users (cost=0.00..11592.75 rows=1 width=0) (actual time=82.826..82.827 rows=0 loops=1)
Filter: ((user_login)::text = 'user_75000'::text)
Buffers: shared read=10660
Planning Time: 0.084 ms
Execution Time: 82.850 ms

Запуск запроса с индексом:

is: multiple query results

SET enable_bitmapscan = on; | -- 1. index user_email_index_hash EXPLA... | -- 2. index user_login_index_hash EXPLAI... | -- 3. index order_type_index_ha

QUERY PLAN
Filter...
Aggregate (cost=8.02..8.03 rows=1 width=8) (actual time=0.021..0.021 rows=1 loops=1)
Buffers: shared read=3
-> Index Scan using user_login_index_hash on users (cost=0.00..8.02 rows=1 width=0) (actual time=0.020..0.020 rows=0 loops=1)
Index Cond: ((user_login)::text = 'user_75000'::text)
Buffers: shared read=3
Planning Time: 0.019 ms
Execution Time: 0.026 ms

3) Проверка order_type_index_hash:

Запуск запроса без индекса:

is: multiple query results

›itmapscan = off; | -- 1. not_index user_email_index_hash E... | -- 2. not_index user_login_index_hash EX... | -- 3. not_index order_type_index_hash E...

QUERY PLAN
Filter...
Finalize Aggregate (cost=57515.26..57515.27 rows=1 width=8) (actual time=682.701..684.699 rows=1 loops=1)
Buffers: shared hit=1 read=38740
-> Gather (cost=57515.05..57515.26 rows=2 width=8) (actual time=682.625..684.691 rows=3 loops=1)
Workers Planned: 2
Workers Launched: 2
Buffers: shared hit=1 read=38740
-> Partial Aggregate (cost=56515.05..56515.06 rows=1 width=8) (actual time=680.571..680.572 rows=1 loops=3)
Buffers: shared hit=1 read=38740
-> Parallel Seq Scan on orders (cost=0.00..56497.29 rows=7102 width=0) (actual time=680.564..680.565 rows=0 loops=3)
Filter: (order_type = 'print'::order_type_enum)
Buffers: shared hit=1 read=38740
Planning:
Buffers: shared hit=48 read=19
Planning Time: 0.196 ms
Execution Time: 684.753 ms

Запуск запроса с индексом:

is: multiple query results

bitmapscan = on; | -- 1. index user_email_index_hash EXPLA... | -- 2. index user_login_index_hash EXPLAI... | -- 3. index order_type_index_hash EXPLA...

QUERY PLAN
Filter...
Aggregate (cost=31599.04..31599.05 rows=1 width=8) (actual time=429.811..429.813 rows=1 loops=1)
Buffers: shared hit=4943 read=38848 written=6
-> Bitmap Heap Scan on orders (cost=624.11..31556.42 rows=17047 width=0) (actual time=429.808..429.809 rows=0 loops=1)
Recheck Cond: (order_type = 'print'::order_type_enum)
Heap Blocks: exact=38744
Buffers: shared hit=4943 read=38848 written=6
-> Bitmap Index Scan on order_type_index_hash (cost=0.00..619.85 rows=17047 width=0) (actual time=51.164..51.165 rows=2053502 loops=1)
Index Cond: (order_type = 'print'::order_type_enum)
Buffers: shared hit=4942 read=105
Planning Time: 0.018 ms
Execution Time: 429.992 ms

4) Проверка order_status_index_hash:

Запуск запроса без индекса:

is: multiple query results

-- 4. not_index order_status_index_hash ... | -- 5. not_index vending_point_schedule_... | -- 6. not_index vending_point_unusual_s... | -- 7. not_index fu

QUERY PLAN
Filter...
Finalize Aggregate (cost=57516.72..57516.73 rows=1 width=8) (actual time=675.323..677.773 rows=1 loops=1)
Buffers: shared hit=1 read=38741
-> Gather (cost=57516.50..57516.71 rows=2 width=8) (actual time=675.131..677.757 rows=3 loops=1)
Workers Planned: 2
Workers Launched: 2
Buffers: shared hit=1 read=38741
-> Partial Aggregate (cost=56516.50..56516.51 rows=1 width=8) (actual time=673.353..673.356 rows=1 loops=3)
Buffers: shared hit=1 read=38741
-> Parallel Seq Scan on orders (cost=0.00..56498.75 rows=7102 width=0) (actual time=673.343..673.344 rows=0 loops=3)
Filter: (order_status = 'completed'::order_status_enum)
Buffers: shared hit=1 read=38741
Planning Time: 0.051 ms
Execution Time: 677.798 ms

Запуск запроса с индексом:

is: multiple query results

-- 4. index order_status_index_hash EXP... | -- 5. index vending_point_schedule_day_... | -- 6. index vending_point_unusual_sched... | -- 7. index functic

QUERY PLAN
Filter...
Aggregate (cost=31599.04..31599.05 rows=1 width=8) (actual time=318.224..318.226 rows=1 loops=1)
Buffers: shared hit=2 read=42110 written=5
-> Bitmap Heap Scan on orders (cost=624.11..31556.42 rows=17047 width=0) (actual time=318.219..318.221 rows=0 loops=1)
Recheck Cond: (order_status = 'completed'::order_status_enum)
Heap Blocks: exact=38744
Buffers: shared hit=2 read=42110 written=5
-> Bitmap Index Scan on order_status_index_hash (cost=0.00..619.85 rows=17047 width=0) (actual time=40.163..40.163 rows=1370035 loops=1)
Index Cond: (order_status = 'completed'::order_status_enum)
Buffers: shared hit=1 read=3367
Planning Time: 0.055 ms
Execution Time: 318.558 ms

5) Проверка vending_point_unusual_schedule_date_index_hash:

Запуск запроса без индекса:

is: multiple query results

-- 4. not_index order_status_index_hash ... | -- 5. not_index vending_point_schedule_... | -- 6. not_index vending_point_unusual_s... | -- 7. not_index fu

QUERY PLAN
Filter...
Aggregate (cost=28.14..28.15 rows=1 width=8) (actual time=0.008..0.008 rows=1 loops=1)
-> Seq Scan on vending_point_schedules (cost=0.00..28.12 rows=7 width=0) (actual time=0.006..0.006 rows=0 loops=1)
Filter: (vending_point_schedule_day_week = 'monday'::day_week_enum)
Planning:
Buffers: shared hit=28 read=11
Planning Time: 0.238 ms
Execution Time: 0.022 ms

Запуск запроса с индексом:

is: multiple query results

-- 4. index order_status_index_hash EXP... | -- 5. index vending_point_schedule_day_... | -- 6. index vending_point_unusual_sched... | -- 7. index functic

QUERY PLAN
Filter...
Aggregate (cost=14.22..14.23 rows=1 width=8) (actual time=0.018..0.018 rows=1 loops=1)
Buffers: shared read=2
-> Bitmap Heap Scan on vending_point_schedules (cost=4.05..14.20 rows=7 width=0) (actual time=0.017..0.017 rows=0 loops=1)
Recheck Cond: (vending_point_schedule_day_week = 'monday'::day_week_enum)
Buffers: shared read=2
-> Bitmap Index Scan on vending_point_schedule_day_week_index_hash (cost=0.00..4.05 rows=7 width=0) (actual time=0.016..0.016 rows=0 loo...
Index Cond: (vending_point_schedule_day_week = 'monday'::day_week_enum)
Buffers: shared read=2
Planning:
Buffers: shared read=1
Planning Time: 0.055 ms
Execution Time: 0.031 ms

6) Проверка vending_point_unusual_schedule_index_hash:

Запуск запроса без индекса:

is: multiple query results

-- 4. not_index order_status_index_hash ... | -- 5. not_index vending_point_schedule_... | -- 6. not_index vending_point_unusual_s... | -- 7. not_index fu

QUERY PLAN
Filter...
Aggregate (cost=35.39..35.40 rows=1 width=8) (actual time=0.009..0.009 rows=1 loops=1)
-> Seq Scan on vending_point_unusual_schedules (cost=0.00..35.38 rows=7 width=0) (actual time=0.008..0.008 rows=0 loops=1)
Filter: (vending_point_unusual_schedule_date = (CURRENT_DATE - '10 days'::interval))
Planning:
Buffers: shared hit=40 read=11
Planning Time: 0.151 ms
Execution Time: 0.015 ms

Запуск запроса с индексом:

is: multiple query results

-- 4. index order_status_index_hash EXP... | -- 5. index vending_point_schedule_day_... | -- 6. index vending_point_unusual_sched... | -- 7. index functic

QUERY PLAN
Filter...
Aggregate (cost=35.39..35.40 rows=1 width=8) (actual time=0.002..0.002 rows=1 loops=1)
-> Seq Scan on vending_point_unusual_schedules (cost=0.00..35.38 rows=7 width=0) (actual time=0.001..0.001 rows=0 loops=1)
Filter: (vending_point_unusual_schedule_date = (CURRENT_DATE - '10 days'::interval))
Planning:
Buffers: shared read=1
Planning Time: 0.022 ms
Execution Time: 0.007 ms

7) Проверка function_variant_index_hash:

Запуск запроса без индекса:

is: multiple query results

ıs_index_hash ... | -- 5. not_index vending_point_schedule_... | -- 6. not_index vending_point_unusual_s... | -- 7. not_index function_variant_index_h...

QUERY PLAN
Filter...
Aggregate (cost=2688.31..2688.32 rows=1 width=8) (actual time=18.392..18.393 rows=1 loops=1)
Buffers: shared read=811
-> Seq Scan on function_variants (cost=0.00..2686.44 rows=750 width=0) (actual time=18.387..18.387 rows=0 loops=1)
Filter: (function_variant = 'color_print'::function_variant_enum)
Buffers: shared read=811
Planning:
Buffers: shared hit=35 read=2
Planning Time: 0.068 ms
Execution Time: 18.403 ms

Запуск запроса с индексом:

is: multiple query results

dex_hash EXP... | -- 5. index vending_point_schedule_day_... | -- 6. index vending_point_unusual_sched... | -- 7. index function_variant_index_hash ...

QUERY PLAN
Filter...
Aggregate (cost=865.05..865.06 rows=1 width=8) (actual time=9.213..9.214 rows=1 loops=1)
Buffers: shared read=935
-> Bitmap Heap Scan on function_variants (cost=25.81..863.17 rows=750 width=0) (actual time=9.209..9.210 rows=0 loops=1)
Recheck Cond: (function_variant = 'color_print'::function_variant_enum)
Heap Blocks: exact=811
Buffers: shared read=935
-> Bitmap Index Scan on function_variant_index_hash (cost=0.00..25.62 rows=750 width=0) (actual time=1.179..1.179 rows=50015 loops=1)
Index Cond: (function_variant = 'color_print'::function_variant_enum)
Buffers: shared read=124
Planning Time: 0.012 ms
Execution Time: 9.223 ms

8) Проверка machine_supplies_datetime_index_btree:

Запуск запроса без индекса:

is: multiple query results

-- 8. not_index machine_supplies_dateti... | -- 9. not_index machine_condition_dateti... | DISCARD PLANS; | -- Включаем индексы для второй

QUERY PLAN
Filter...
Finalize Aggregate (cost=40375.37..40375.38 rows=1 width=8) (actual time=419.437..421.489 rows=1 loops=1)
Buffers: shared read=15924
-> Gather (cost=40375.16..40375.37 rows=2 width=8) (actual time=419.357..421.481 rows=3 loops=1)
Workers Planned: 2
Workers Launched: 2
Buffers: shared read=15924
-> Partial Aggregate (cost=39375.16..39375.17 rows=1 width=8) (actual time=416.864..416.865 rows=1 loops=3)
Buffers: shared read=15924
-> Parallel Seq Scan on machine_supplies (cost=0.00..39362.14 rows=5208 width=0) (actual time=416.854..416.854 rows=0 loops=3)
Filter: ((machine_supplies_datetime <= now()) AND (machine_supplies_datetime >= (now() - '30 days'::interval)))
Buffers: shared read=15924
Planning:
Buffers: shared hit=43 read=8
Planning Time: 0.198 ms
Execution Time: 421.508 ms

Запуск запроса с индексом:

is: multiple query results

unusual_sched... | -- 7. index function_variant_index_hash ... | -- 8. index machine_supplies_datetime_i... | -- 9. index machine_condition_datetime_...

QUERY PLAN
Filter...
Aggregate (cost=16312.56..16312.57 rows=1 width=8) (actual time=74.021..74.023 rows=1 loops=1)
Buffers: shared read=16284 written=4
-> Bitmap Heap Scan on machine_supplies (cost=312.56..16281.31 rows=12500 width=0) (actual time=74.017..74.018 rows=0 loops=1)
Recheck Cond: ((machine_supplies_datetime >= (now() - '30 days'::interval)) AND (machine_supplies_datetime <= now()))
Heap Blocks: exact=15902
Buffers: shared read=16284 written=4
-> Bitmap Index Scan on machine_supplies_datetime_index_btree (cost=0.00..309.44 rows=12500 width=0) (actual time=12.546..12.547 rows=10...
Index Cond: ((machine_supplies_datetime >= (now() - '30 days'::interval)) AND (machine_supplies_datetime <= now()))
Buffers: shared read=382
Planning Time: 0.058 ms
Execution Time: 74.044 ms

9) Проверка machine_condition_datetime_index_btree:

Запуск запроса без индекса:

is: multiple query results

-- 8. not_index machine_supplies_dateti... | -- 9. not_index machine_condition_dateti... | DISCARD PLANS; | -- Включаем индексы для второй

QUERY PLAN
Filter...
Finalize Aggregate (cost=34053.28..34053.29 rows=1 width=8) (actual time=341.289..343.293 rows=1 loops=1)
Buffers: shared read=12739
-> Gather (cost=34053.06..34053.27 rows=2 width=8) (actual time=341.213..343.287 rows=3 loops=1)
Workers Planned: 2
Workers Launched: 2
Buffers: shared read=12739
-> Partial Aggregate (cost=33053.06..33053.07 rows=1 width=8) (actual time=339.263..339.264 rows=1 loops=3)
Buffers: shared read=12739
-> Parallel Seq Scan on machine_conditions (cost=0.00..33041.78 rows=4512 width=0) (actual time=339.253..339.254 rows=0 loops=3)
Filter: ((machine_condition_datetime <= now()) AND (machine_condition_datetime >= (now() - '7 days'::interval)))
Buffers: shared read=12739
Planning:
Buffers: shared hit=26 read=15
Planning Time: 0.209 ms
Execution Time: 343.312 ms

Запуск запроса с индексом:

is: multiple query results

unusual_sched... | -- 7. index function_variant_index_hash ... | -- 8. index machine_supplies_datetime_i... | -- 9. index machine_condition_datetime_...

QUERY PLAN
Filter...
Aggregate (cost=13323.01..13323.02 rows=1 width=8) (actual time=33.547..33.549 rows=1 loops=1)
Buffers: shared read=9968
-> Bitmap Heap Scan on machine_conditions (cost=263.42..13295.94 rows=10828 width=0) (actual time=33.543..33.544 rows=0 loops=1)
Recheck Cond: ((machine_condition_datetime >= (now() - '7 days'::interval)) AND (machine_condition_datetime <= now()))
Heap Blocks: exact=9901
Buffers: shared read=9968
-> Bitmap Index Scan on machine_condition_datetime_index_btree (cost=0.00..260.72 rows=10828 width=0) (actual time=2.275..2.275 rows=189...
Index Cond: ((machine_condition_datetime >= (now() - '7 days'::interval)) AND (machine_condition_datetime <= now()))
Buffers: shared read=67
Planning Time: 0.061 ms
Execution Time: 33.566 ms

Таблица сравнения выполнения запросов без индексов и с индексами:

№	С индексами или без индексов	Execution Time (ms)	Разница Execution Time (ms)	Разница Execution Time (%)
1	Без индекса	82,915	82,861	99,87%
	С индексом	0,054		
2	Без индекса	82,850	82,824	99,94%
	С индексом	0,026		
3	Без индекса	684,753	254,761	22,85%
	С индексом	429,992		
4	Без индекса	677,798	359,24	36,06%
	С индексом	318,558		
5	Без индекса	0,022	-0,009	-16,98%
	С индексом	0,031		
6	Без индекса	0,015	0,008	36,36%
	С индексом	0,007		
7	Без индекса	18,403	9,18	33,23%
	С индексом	9,223		
8	Без индекса	421,508	347,464	70,12%
	С индексом	74,044		
9	Без индекса	343,312	309,746	82,19%
	С индексом	33,566		

На основе представленных результатов и проведения дополнительных тестов, было принято решение удалить индекс для поля vending_point_schedule_day_week таблицы vending_point_schedules (5 индекс).
Также были поменяны типы индексов для полей order_type и order_status таблицы orders (индексы 3 и 4) и для поля function_variant таблицы function_variants (индекс 7) с hash на btree.

Список индексов после оптимизации:

```
CREATE INDEX user_email_index_hash ON users USING hash(user_email);
CREATE INDEX user_login_index_hash ON users USING hash(user_login);

CREATE INDEX order_type_index_btree ON orders USING btree(order_type);
CREATE INDEX order_status_index_btree ON orders USING btree(order_status);

CREATE INDEX vending_point_unusual_schedule_date_index_hash ON vending_point_unusual_schedules
   USING hash(vending_point_unusual_schedule_date);

CREATE INDEX function_variant_index_btree ON function_variants USING btree(function_variant);

CREATE INDEX machine_supplies_datetime_index_btree ON machine_supplies USING
btree(machine_supplies_datetime);
CREATE INDEX machine_condition_datetime_index_btree ON machine_conditions USING
btree(machine_condition_datetime);
```

Результаты тестов, после проведенной оптимизации:

Размер бд – 1,4 Гб (1439 Мб)

check_indexes.sql | script_generate_data.sql | create_indexes.sql | check_db_size.sql | is-coursework: -- Размер базы д...

check_db_size.sql

Run on active connection | Select block

```
-- Размер базы данных
SELECT pg_size_pretty(pg_database_size(current_database())) as database_
```

database_size_...
1439 MB

Выполнение скрипта проверки:

1) Проверка user_email_index_hash:

Запуск запроса без индекса:

is-coursework: multiple query results

рование индексов - ... | SET enable_bitmapscan = off; | -- 1. not_index user_email_index_hash E...

QUERY PLAN
Aggregate (cost=15305.94..15305.95 rows=1 width=8) (actual time=19.643..21.892 rows=1 loop...
Buffers: shared hit=32 read=10628
-> Gather (cost=1000.00..15305.93 rows=1 width=0) (actual time=2.962..21.883 rows=1 loops=1)
Workers Planned: 2
Workers Launched: 2
Buffers: shared hit=32 read=10628
-> Parallel Seq Scan on users (cost=0.00..14305.83 rows=1 width=0) (actual time=12.442..17.982 ...
Filter: ((user_email)::text = 'user_50000@example.com'::text)
Rows Removed by Filter: 233333
Buffers: shared hit=32 read=10628
Planning:
Buffers: shared hit=128 read=2
Planning Time: 0.274 ms
Execution Time: 21.932 ms

Запуск запроса с индексом:

is-coursework: multiple query results

зторой части... | SET enable_bitmapscan = on; | -- 1. index user_email_index_hash EXPLA... | -- 2. in

QUERY PLAN
Filter...
Aggregate (cost=4.45..4.46 rows=1 width=8) (actual time=0.053..0.053 rows=1 loops=1)
Buffers: shared hit=1 read=3
-> Index Only Scan using users_user_email_key on users (cost=0.42..4.44 rows=1 width=0) (actua…
Index Cond: (user_email = 'user_50000@example.com'::text)
Heap Fetches: 0
Buffers: shared hit=1 read=3
Planning Time: 0.052 ms
Execution Time: 0.069 ms

2) Проверка user_login_index_hash:

Запуск запроса без индекса:

is-coursework: multiple query results

-- 2. not_index user_login_index_hash EX... | -- 3. not_index order_type_index_btree E... | -- 4. not_i

QUERY PLAN
Filter...
Aggregate (cost=15305.94..15305.95 rows=1 width=8) (actual time=20.727..22.491 rows=1 loop…
Buffers: shared hit=128 read=10532
-> Gather (cost=1000.00..15305.93 rows=1 width=0) (actual time=3.776..22.484 rows=1 loops=1)
Workers Planned: 2
Workers Launched: 2
Buffers: shared hit=128 read=10532
-> Parallel Seq Scan on users (cost=0.00..14305.83 rows=1 width=0) (actual time=13.494..19.124 …
Filter: ((user_login)::text = 'user_75000'::text)
Rows Removed by Filter: 233333
Buffers: shared hit=128 read=10532
Planning:
Buffers: shared hit=3
Planning Time: 0.091 ms
Execution Time: 22.509 ms

Запуск запроса с индексом:

is-coursework: multiple query results

;can = on; | -- 1. index user_email_index_hash EXPLA... | -- 2. index user_login_index_hash EXPLAI...

QUERY PLAN
Filter...
Aggregate (cost=4.45..4.46 rows=1 width=8) (actual time=0.021..0.022 rows=1 loops=1)
Buffers: shared hit=1 read=3
-> Index Only Scan using users_user_login_key on users (cost=0.42..4.44 rows=1 width=0) (actual...
Index Cond: (user_login = 'user_75000'::text)
Heap Fetches: 0
Buffers: shared hit=1 read=3
Planning Time: 0.019 ms
Execution Time: 0.027 ms

3) Проверка order_type_index_hash:

Запуск запроса без индекса:

is-coursework: multiple query results

-- 2. not_index user_login_index_hash EX... | -- 3. not_index order_type_index_btree E... | -- 4. not_i

QUERY PLAN
Filter...
Finalize Aggregate (cost=76828.45..76828.46 rows=1 width=8) (actual time=174.431..176.429 ro...
Buffers: shared hit=128 read=47052
-> Gather (cost=76828.23..76828.44 rows=2 width=8) (actual time=174.360..176.422 rows=3 loo...
Workers Planned: 2
Workers Launched: 2
Buffers: shared hit=128 read=47052
-> Partial Aggregate (cost=75828.23..75828.24 rows=1 width=8) (actual time=172.883..172.884 r...
Buffers: shared hit=128 read=47052
-> Parallel Seq Scan on orders (cost=0.00..73221.17 rows=1042827 width=0) (actual time=0.013.....
Filter: (order_type = 'print'::order_type_enum)
Rows Removed by Filter: 833794
Buffers: shared hit=128 read=47052
Planning:
Buffers: shared hit=56
Planning Time: 0.130 ms
Execution Time: 176.454 ms

Запуск запроса с индексом:

is-coursework: multiple query results

-- 3. index order_type_index_btree EXPL... | -- 4. index order_status_index_btree EXP... | -- 5. index

QUERY PLAN
Filter...
Finalize Aggregate (cost=41282.87..41282.88 rows=1 width=8) (actual time=110.327..112.644 ro...
Buffers: shared hit=6 read=2106
-> Gather (cost=41282.66..41282.87 rows=2 width=8) (actual time=110.321..112.639 rows=3 loo...
Workers Planned: 2
Workers Launched: 2
Buffers: shared hit=6 read=2106
-> Partial Aggregate (cost=40282.66..40282.67 rows=1 width=8) (actual time=108.669..108.670 r...
Buffers: shared hit=6 read=2106
-> Parallel Index Only Scan using order_type_index_btree on orders (cost=0.43..37675.59 rows=...
Index Cond: (order_type = 'print'::order_type_enum)
Heap Fetches: 0
Buffers: shared hit=6 read=2106
Planning Time: 0.029 ms
Execution Time: 112.670 ms

4) Проверка order_status_index_hash:

Запуск запроса без индекса:

is-coursework: multiple query results

ash EX... | -- 3. not_index order_type_index_btree E... | -- 4. not_index order_status_index_btree...

QUERY PLAN
Filter...
Finalize Aggregate (cost=75939.23..75939.24 rows=1 width=8) (actual time=160.133..161.933 ro...
Buffers: shared hit=224 read=46956
-> Gather (cost=75939.02..75939.23 rows=2 width=8) (actual time=160.072..161.927 rows=3 loo...
Workers Planned: 2
Workers Launched: 2
Buffers: shared hit=224 read=46956
-> Partial Aggregate (cost=74939.02..74939.03 rows=1 width=8) (actual time=158.236..158.237 r...
Buffers: shared hit=224 read=46956
-> Parallel Seq Scan on orders (cost=0.00..73221.17 rows=687140 width=0) (actual time=0.017..1...
Filter: (order_status = 'completed'::order_status_enum)
Rows Removed by Filter: 1111468
Buffers: shared hit=224 read=46956
Planning:
Buffers: shared hit=3
Planning Time: 0.091 ms
Execution Time: 161.956 ms

Запуск запроса с индексом:

is-coursework: multiple query results

-- 3. index order_type_index_btree EXPL... | -- 4. index order_status_index_btree EXP... | -- 5. inde›

QUERY PLAN
Filter...
Finalize Aggregate (cost=27542.41..27542.42 rows=1 width=8) (actual time=74.197..76.236 rows...
Buffers: shared hit=6 read=1405 written=1070
-> Gather (cost=27542.19..27542.40 rows=2 width=8) (actual time=74.154..76.230 rows=3 loops=1)
Workers Planned: 2
Workers Launched: 2
Buffers: shared hit=6 read=1405 written=1070
-> Partial Aggregate (cost=26542.19..26542.20 rows=1 width=8) (actual time=72.367..72.367 ro...
Buffers: shared hit=6 read=1405 written=1070
-> Parallel Index Only Scan using order_status_index_btree on orders (cost=0.43..24824.34 rows...
Index Cond: (order_status = 'completed'::order_status_enum)
Heap Fetches: 0
Buffers: shared hit=6 read=1405 written=1070
Planning Time: 0.066 ms
Execution Time: 76.254 ms

5) Проверка vending_point_unusual_schedule_date_index_hash:

Запуск запроса без индекса:

is-coursework: multiple query results

-- 5. not_index vending_point_unusual_s... | -- 6. not_index function_variant_index_b... | -- 7. not_i

QUERY PLAN
Filter...
Aggregate (cost=35.39..35.40 rows=1 width=8) (actual time=0.011..0.012 rows=1 loops=1)
-> Seq Scan on vending_point_unusual_schedules (cost=0.00..35.38 rows=7 width=0) (actual tim...
Filter: (vending_point_unusual_schedule_date = (CURRENT_DATE - '10 days'::interval))
Planning:
Buffers: shared hit=50 read=1
Planning Time: 0.140 ms
Execution Time: 0.021 ms

Запуск запроса с индексом:

is-coursework: multiple query results

e EXPL... | -- 4. index order_status_index_btree EXP... | -- 5. index vending_point_unusual_sched...

QUERY PLAN
Filter...
Aggregate (cost=35.39..35.40 rows=1 width=8) (actual time=0.003..0.003 rows=1 loops=1)
-> Seq Scan on vending_point_unusual_schedules (cost=0.00..35.38 rows=7 width=0) (actual tim...
Filter: (vending_point_unusual_schedule_date = (CURRENT_DATE - '10 days'::interval))
Planning:
Buffers: shared hit=1
Planning Time: 0.052 ms
Execution Time: 0.012 ms

6) Проверка function_variant_index_hash:

Запуск запроса без индекса:

is-coursework: multiple query results

-- 5. not_index vending_point_unusual_s... | -- 6. not_index function_variant_index_b... | -- 7. not_i

QUERY PLAN
Filter...
Aggregate (cost=2811.12..2811.14 rows=1 width=8) (actual time=11.561..11.562 rows=1 loops=1)
Buffers: shared hit=811
-> Seq Scan on function_variants (cost=0.00..2686.00 rows=50050 width=0) (actual time=0.009.....
Filter: (function_variant = 'color_print'::function_variant_enum)
Rows Removed by Filter: 99937
Buffers: shared hit=811
Planning:
Buffers: shared hit=33
Planning Time: 0.076 ms
Execution Time: 11.570 ms

Запуск запроса с индексом:

is-coursework: multiple query results

-- 6. index function_variant_index_btree ... | -- 7. index machine_supplies_datetime_i... | -- 8. inde›

QUERY PLAN
Filter...
Aggregate (cost=1177.30..1177.31 rows=1 width=8) (actual time=6.655..6.656 rows=1 loops=1)
Buffers: shared hit=1 read=44 written=44
-> Index Only Scan using function_variant_index_btree on function_variants (cost=0.29..1052.17 ...
Index Cond: (function_variant = 'color_print'::function_variant_enum)
Heap Fetches: 0
Buffers: shared hit=1 read=44 written=44
Planning Time: 0.028 ms
Execution Time: 6.667 ms

7) Проверка machine_supplies_datetime_index_btree:

Запуск запроса без индекса:

is-coursework: multiple query results

sual_s... | -- 6. not_index function_variant_index_b... | -- 7. not_index machine_supplies_dateti...

QUERY PLAN
Filter...
Finalize Aggregate (cost=40469.78..40469.79 rows=1 width=8) (actual time=185.902..187.769 ro...
Buffers: shared hit=64 read=15860
-> Gather (cost=40469.57..40469.78 rows=2 width=8) (actual time=185.822..187.764 rows=3 loo...
Workers Planned: 2
Workers Launched: 2
Buffers: shared hit=64 read=15860
-> Partial Aggregate (cost=39469.57..39469.58 rows=1 width=8) (actual time=184.312..184.312 r...
Buffers: shared hit=64 read=15860
-> Parallel Seq Scan on machine_supplies (cost=0.00..39361.50 rows=43228 width=0) (actual tim...
Filter: ((machine_supplies_datetime <= now()) AND (machine_supplies_datetime >= (now() - '30 ...
Rows Removed by Filter: 799142
Buffers: shared hit=64 read=15860
Planning:
Buffers: shared hit=52 read=3
Planning Time: 0.232 ms
Execution Time: 187.788 ms

Запуск запроса с индексом:

is-coursework: multiple query results

-- 6. index function_variant_index_btree ... | -- 7. index machine_supplies_datetime_i... | -- 8. inde›

QUERY PLAN
Filter...
Aggregate (cost=3474.74..3474.75 rows=1 width=8) (actual time=22.719..22.720 rows=1 loops=1)
Buffers: shared hit=6 read=281 written=281
-> Index Only Scan using machine_supplies_datetime_index_btree on machine_supplies (cost=0....
Index Cond: ((machine_supplies_datetime >= (now() - '30 days'::interval)) AND (machine_supplie...
Heap Fetches: 0
Buffers: shared hit=6 read=281 written=281
Planning:
Buffers: shared hit=4
Planning Time: 0.081 ms
Execution Time: 22.733 ms

8) Проверка machine_condition_datetime_index_btree:

Запуск запроса без индекса:

is-coursework: multiple query results

-- 8. not_index machine_condition_dateti... | DISCARD PLANS; | -- Включаем индексы для в

QUERY PLAN
Filter...
Finalize Aggregate (cost=32510.61..32510.62 rows=1 width=8) (actual time=141.992..143.806 ro...
Buffers: shared hit=64 read=12675
-> Gather (cost=32510.40..32510.61 rows=2 width=8) (actual time=141.864..143.799 rows=3 loo...
Workers Planned: 2
Workers Launched: 2
Buffers: shared hit=64 read=12675
-> Partial Aggregate (cost=31510.40..31510.41 rows=1 width=8) (actual time=140.315..140.316 r...
Buffers: shared hit=64 read=12675
-> Parallel Seq Scan on machine_conditions (cost=0.00..31489.00 rows=8559 width=0) (actual ti...
Filter: ((machine_condition_datetime <= now()) AND (machine_condition_datetime >= (now() - '...
Rows Removed by Filter: 660283
Buffers: shared hit=64 read=12675
Planning:
Buffers: shared hit=38 read=3
Planning Time: 0.162 ms
Execution Time: 143.829 ms

Запуск запроса с индексом:

is-coursework: multiple query results

_btree ... | -- 7. index machine_supplies_datetime_i... | -- 8. index machine_condition_datetime_...

QUERY PLAN

Filter...

```
Aggregate (cost=690.61..690.62 rows=1 width=8) (actual time=4.051..4.052 rows=1 loops=1)
Buffers: shared hit=4 read=52 written=52
-> Index Only Scan using machine_condition_datetime_index_btree on machine_conditions (cost...
Index Cond: ((machine_condition_datetime >= (now() - '7 days'::interval)) AND (machine_conditi...
Heap Fetches: 0
Buffers: shared hit=4 read=52 written=52
Planning:
Buffers: shared hit=8
Planning Time: 0.072 ms
Execution Time: 4.069 ms
```

Таблица сравнения выполнения запросов без индексов и с индексами:

№	С индексами или без индексов	Execution Time (ms)	Разница Execution Time (ms)	Разница Execution Time (%)
1	Без индекса	21,932	21,863	99,69%
	С индексом	0,069		
2	Без индекса	22,509	22,482	99,88%
	С индексом	0,027		
3	Без индекса	176,454	63,784	36,15%
	С индексом	112,670		
4	Без индекса	161,956	85,702	52,92%
	С индексом	76,254		
5	Без индекса	0,021	0,009	42,86%
	С индексом	0,012		
6	Без индекса	11,570	4,903	42,38%
	С индексом	6,667		
7	Без индекса	187,788	165,055	87,89%
	С индексом	22,733		
8	Без индекса	143,829	139,76	97,17%
	С индексом	4,069		

Вывод по использованию индексов:

По планам выполнения запросов видно, то, что индексы используются. При этом все создаваемые индексы значительно оптимизируют время выполнения запросов, поэтому их использование оправдано.